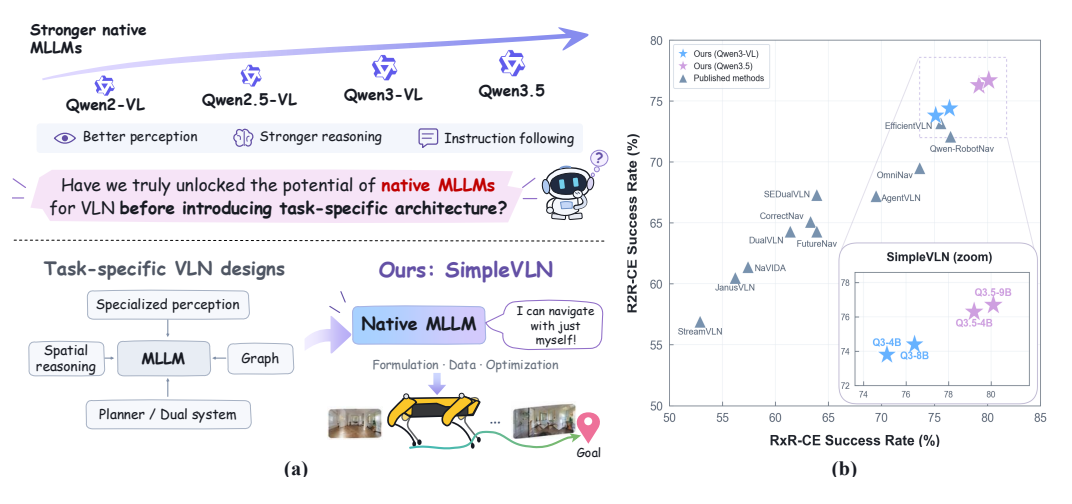

Stronger native MLLMs
Qwen2-VL
Qwen2.5-VL
Qwen3-VL
Qwen3.5
Better perception
Stronger reasoning
Instruction following
Have we truly unlocked the potential of native MLLMs for VLN before introducing task-specific architecture?
Task-specific VLN designs
Specialized perception
Spatial reasoning
MLLM
Graph
Planner / Dual system
Ours: SimpleVLN
Native MLLM
I can navigate with just myself!
Formulation · Data · Optimization
Goal
(a)
Ours (Qwen3-VL)
Ours (Qwen3.5)
Published methods
EfficientVLN
Qwen-RobotNav
OmniNav
AgentVLN
SEDualVLN
CorrectNav
DualVLN
FutureNav
NaVIDA
JanusVLN
StreamVLN
SimpleVLN (zoom)
Q3.5-9B
Q3.5-4B
Q3-4B
Q3-8B
R2R-CE Success Rate (%)
RxR-CE Success Rate (%)
(b)